

Paired 't' test for difference of means

The test is used when,

- i) sample sizes are equal ($n_1 = n_2$)
- ii) Sample observations $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ are not completely independent.

For example, if we want to find out an advertisement is really effective in promoting sales of a particular product, let $x_1, x_2, \dots, x_n$ be the sales of the product in 'n' departmental stores for a period of time before the advertisement campaign. Let $y_1, y_2, \dots, y_n$ be the corresponding sales of same product after advertisement campaign in the same departmental stores. Let (x_i, y_i) be the sales in the i^{th} department store. In this case we apply paired 't' test as given below.

Let $d_i = x_i - y_i$ be the difference in the observations for the i^{th} unit.

Null hypothesis, $H_0: \mu_x = \mu_y$

$$\text{Test statistic, } t = \frac{\bar{d}}{\sqrt{s^2/n}}$$

$$\text{where } d_i = x_i - y_i \text{ \& } \bar{d} = \frac{\sum d_i}{n}$$

$$s^2 = \frac{1}{(n-1)} \sum (d_i - \bar{d})^2$$

the test statistic follows a 't' distribution with $(n-1)$ degrees of freedom.

ex: An IQ Test was given to 5 people before and after they were trained. The results are given as follows:

Candidates:	1	2	3	4	5
IQ before:	110	120	123	132	125
IQ after:	120	118	125	136	121

Test whether there is any change after the training program.

$$H_0: \mu_x = \mu_y$$

$$H_1: \mu_x \neq \mu_y$$

Level of significance = 5% = 0.05

$$t = \frac{\bar{d}}{\sqrt{s^2/n}}$$

x_i	y_i	$d_i = x_i - y_i$	$d_i - \bar{d}$	$(d_i - \bar{d})^2$
110	120	-10	8	64
120	118	2	4	16
123	125	-2	0	0
132	136	-4	-2	4
125	121	$\frac{4}{-10}$	$\frac{6}{-10}$	$\frac{36}{120}$

$$\bar{d} = \frac{\sum d_i}{n} = \frac{-10}{5} = \underline{\underline{-2}}$$

$$s^2 = \frac{1}{(n-1)} \sum (d_i - \bar{d})^2 = \frac{120}{4} = \underline{\underline{30}}$$

$$\text{Test statistic, } t = \frac{\bar{d}}{\sqrt{s^2/n}}$$

$$= \frac{-2}{\sqrt{30/5}}$$

$$= \frac{-2}{\sqrt{6}} = \underline{\underline{-0.816}}$$

$$|t| = 0.816$$

Table value, $t(0.05, 4) = 2.776$

Calculated value $<$ table value

$\therefore H_0$ is accepted.

Hence we can conclude that there is no change in the training programs.

Ques: The sales data of an item in 6 shops before and after a special campaign are as follows;

shops :	A	B	C	D	E	F
Before :	53	28	31	48	50	42
After :	58	29	30	55	56	45

Can the campaign be judged to be a success.
Test at 5% level of significance.

$$H_0 : \mu_x = \mu_y$$

$$H_1 : \mu_x < \mu_y$$

Level of significance = 5% = 0.05.

x_i	y_i	$d_i = x_i - y_i$	$d_i - \bar{d}$	$(d_i - \bar{d})^2$
53	58	-5	-15	2.25
28	29	-1	-2.5	6.25
31	30	1	4.5	20.25
48	55	-7	-3.5	12.25
50	56	-6	-2.5	6.25
42	45	-3	0.5	0.25
				<u>47.5</u>

$$\bar{d} = \frac{\sum d_i}{n} = \frac{-21}{6} = -3.5$$

$$s^2 = \frac{1}{n-1} \sum (d_i - \bar{d})^2$$

$$= \frac{47.5}{5} = \underline{\underline{9.5}}$$

$$t = \frac{\bar{d}}{\sqrt{s^2/n}}$$

$$t = \frac{-3.5}{\sqrt{9.5/6}} = \underline{\underline{-2.8}}$$

$$|t| = 2.8$$

From table, $t(0.05, 5) = 2.015$

Calculated value $>$ table value.

$\therefore H_0$ is rejected.

ie H_1 is accepted.

$\therefore$ Hence the campaign is a
success